Study notes: Module 3, electromagnetic waves

Griffiths, *Introduction to Electrodynamics* (5th ed.), §8.1.2 and Chapter 9. Figures are redrawn for these notes, not copied from the book; the book figure number is in each caption. Sections 4 (§9.3.3) and 5 (§9.4, §9.5) are background, beyond the syllabus list. Problems are not included.

Section 1: energy and Poynting's theorem (Griffiths 8.1.2)

8.1.2 Poynting's theorem

Source: book-sources/m3-s812-poynting.html, Griffiths pp. 363–367. Equations 8.5–8.12 and Example 8.1 are included in full. Problems 8.1–8.4 are not included.

Before you start

- The divergence theorem: $\oint_S \mathbf{A}\cdot d\mathbf{a} = \int_\mathcal{V}(\nabla\cdot\mathbf{A})\,d\tau$ (Beyond in Plus Two; see pre-math M13).
- The product rule for a divergence: $\nabla\cdot(\mathbf{E}\times\mathbf{B}) = \mathbf{B}\cdot(\nabla\times\mathbf{E}) - \mathbf{E}\cdot(\nabla\times\mathbf{B})$.
- Faraday's law and the Ampère–Maxwell law, in the forms given in Module 2 (Section 7.3).

1. The energy stored in the fields

Earlier in the book, two results were found for the work needed to build up a field. To assemble a static charge distribution, the work is $W_e = \frac{\epsilon_0}{2}\int E^2\,d\tau$ (Eq. 2.45). To get currents flowing against the back emf, the work is $W_m = \frac{1}{2\mu_0}\int B^2\,d\tau$ (Eq. 7.44). Both are integrals of a density, so the energy per unit volume must be:

$$u = \frac{1}{2}\left(\epsilon_0 E^2 + \frac{1}{\mu_0}B^2\right) \tag{8.5}$$

Section 8.1.2 checks this, and then builds the energy conservation law from it.

2. Work done on the charges

Take a charge and current distribution that makes fields $\mathbf{E}$ and $\mathbf{B}$ at time t. In the next instant dt the charges move a little. The Lorentz force does work on each charge. The magnetic part does no work, because $\mathbf{v}\times\mathbf{B}$ is at right angles to $\mathbf{v}$. So the work on a charge q is

$$dW = q\mathbf{E}\cdot\mathbf{v}\,dt$$

To pass from one charge to the whole volume, replace q by $\rho\,d\tau$ and $\rho\mathbf{v}$ by $\mathbf{J}$. The rate at which work is done on all the charges in $\mathcal{V}$ is then

$$\frac{dW}{dt} = \int_\mathcal{V}(\mathbf{E}\cdot\mathbf{J})\,d\tau \tag{8.6}$$

The quantity $\mathbf{E}\cdot\mathbf{J}$ is the power per unit volume that the fields deliver to the charges.

3. Eliminating J

Write $\mathbf{J}$ in terms of the fields, using the Ampère–Maxwell law. That gives

$$\mathbf{E}\cdot\mathbf{J} = \frac{1}{\mu_0}\mathbf{E}\cdot(\nabla\times\mathbf{B}) - \epsilon_0\,\mathbf{E}\cdot\frac{\partial\mathbf{E}}{\partial t}$$

The first term still contains $\nabla\times\mathbf{B}$. Use the product rule, and then Faraday's law, to rewrite it:

$$\mathbf{E}\cdot(\nabla\times\mathbf{B}) = -\mathbf{B}\cdot\frac{\partial\mathbf{B}}{\partial t} - \nabla\cdot(\mathbf{E}\times\mathbf{B})$$

Two of the terms now have the form of a time derivative. Since $\mathbf{B}\cdot\partial\mathbf{B}/\partial t = \frac{1}{2}\partial(B^2)/\partial t$, and similarly for $\mathbf{E}$:

$$\mathbf{B}\cdot\frac{\partial\mathbf{B}}{\partial t} = \frac{1}{2}\frac{\partial}{\partial t}(B^2) \qquad \mathbf{E}\cdot\frac{\partial\mathbf{E}}{\partial t} = \frac{1}{2}\frac{\partial}{\partial t}(E^2) \tag{8.7}$$

Putting these together:

$$\mathbf{E}\cdot\mathbf{J} = -\frac{1}{2}\frac{\partial}{\partial t}\left(\epsilon_0 E^2 + \frac{1}{\mu_0}B^2\right) - \frac{1}{\mu_0}\nabla\cdot(\mathbf{E}\times\mathbf{B}) \tag{8.8}$$

The first term is minus the time rate of change of the energy density u (Eq. 8.5). The second term is a divergence. So part of the power delivered to the

charges comes from a fall in the stored energy, and part comes from a divergence term that describes flow through the surface.

4. Poynting's theorem

Substitute Eq. 8.8 into Eq. 8.6. Apply the divergence theorem to the second term, which turns the volume integral into a surface integral over the boundary S of $\mathcal{V}$:

$$\frac{dW}{dt} = -\frac{d}{dt}\int_{\mathcal{V}} u\, d\tau - \frac{1}{\mu_0}\oint_S (\mathbf{E}\times\mathbf{B})\cdot d\mathbf{a} \tag{8.9}$$

In words: the work done on the charges equals the fall in energy stored in the fields, minus the energy that flows out through the surface. This is Poynting's theorem. It is the work–energy theorem for electrodynamics.

> **PLUS TWO RECALL: POWER IN A CIRCUIT**
>
> **Recall.** From Class 12: the power delivered to a resistor is $P = VI = I^2R$, the rate of Joule heating.
>
> **Re-derive.**
>
> 1. Power is energy per unit time. For a resistor, the energy per unit time that becomes heat is the work done on the charges per unit time.
> 2. Using $V = IR$, $VI = I^2R$ is the same quantity written in two ways.
> 3. So the power of Poynting's theorem, which the fields deliver to the charges in a wire, must reproduce VI when we check it with a real wire (Example 8.1 below).
>
> **Saved.** $P = VI$. This is the result that Example 8.1 reproduces from the Poynting vector.

5. The Poynting vector

The surface integral in Eq. 8.9 is the energy that leaves the volume per unit time. Define the vector

$$\mathbf{S} \equiv \frac{1}{\mu_0}(\mathbf{E}\times\mathbf{B}) \tag{8.10}$$

$\mathbf{S}$ is the **Poynting vector**. It is the energy per unit time crossing unit area: the energy flux density. Then $\mathbf{S}\cdot d\mathbf{a}$ is the energy crossing an element of area $d\mathbf{a}$ in unit time.

Two cautions from the book (footnote 4). First, Eq. 8.9 shows that $\oint \mathbf{S}\cdot d\mathbf{a}$ is the power through a *closed* surface. It does not prove that $\int \mathbf{S}\cdot d\mathbf{a}$ is the power through an *open* surface, because a term could integrate to zero over every closed surface. Second, where energy is located is not fixed in electrodynamics (Section 2.4.4). The Poynting vector is the natural interpretation, not a proof of location.

With Eq. 8.10, Poynting's theorem takes a compact form:

$$\frac{dW}{dt} = -\frac{d}{dt}\int_{\mathcal{V}} u\, d\tau - \oint_S \mathbf{S}\cdot d\mathbf{a} \tag{8.11}$$

6. Local conservation of energy

Take a region with no charge, so $dW/dt = 0$. Then Eq. 8.11 gives $\partial_t \int u\, d\tau = -\oint \mathbf{S}\cdot d\mathbf{a}$. Apply the divergence theorem to the right side, and then, because the volume is arbitrary, the integrands must match:

> **RESULT 8.1: LOCAL ENERGY CONSERVATION IN EMPTY SPACE**
>
> $$\frac{\partial u}{\partial t} = -\nabla\cdot\mathbf{S} \tag{8.12}$$
>
> This has the same form as the continuity equation for charge (Eq. 8.4). u plays the part of ρ, and $\mathbf{S}$ plays the part of $\mathbf{J}$.

In general, electromagnetic energy by itself is not conserved. The fields do work on the charges, and the charges make fields. Energy moves back and forth between them, so the total energy of matter plus fields is what is conserved.

7. Example 8.1: a resistive wire

Current flows down a resistive wire of length L and radius a. The book uses the Poynting vector to find the power delivered to the wire, and checks it against VI.

Step 1. The potential difference between the ends is V. The field inside the wire is uniform and points along the wire: $E = V/L$.

Step 2. The current I produces a circumferential magnetic field. At the surface, $B = \mu_0 I/2\pi a$ (Ampère's law).

Step 3. The two fields are at right angles. So the magnitude of the Poynting vector is

$$S = \frac{1}{\mu_0} \cdot \frac{V}{L} \cdot \frac{\mu_0 I}{2\pi a} = \frac{VI}{2\pi a L}$$

Step 4. $\mathbf{E}$ runs along the wire and $\mathbf{B}$ goes round it, so $\mathbf{E} \times \mathbf{B}$ points radially. Its direction is radially inward, toward the axis of the wire.

Step 5. The surface of the wire has area $2\pi a L$. The energy per unit time crossing it inward is

$$\int \mathbf{S} \cdot d\mathbf{a} = S(2\pi a L) = VI$$

This is exactly the Joule heating VI found in Section 7.1.1. The energy comes into the wire through its surface, from the fields outside it, rather than along the wire.

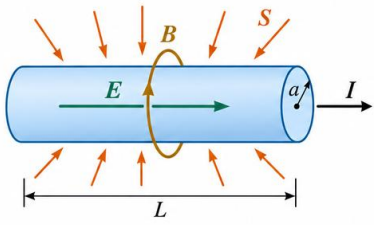


Fig. 8.1 (book): a resistive wire carrying current I; E along the wire, B circumferential, S radially inward.

Note the scope: the book's Problems 8.1–8.4 (cables, a capacitor gap, a charging sphere) apply the same theorem and are not included in these notes.

Section 2: waves in one dimension (Griffiths 9.1)

9.1.1–9.1.2 The wave equation and sinusoidal waves

Source: book-sources/m3-sec2-wave1d.html, Griffiths pp. 390–396. Equations 9.1–9.20 are included in full. Problems 9.1–9.4 are listed but not worked.

Before you start

- Partial derivatives and the chain rule (pre-math M7): if f depends on $z - vt$, then $\partial f/\partial t = -v\,\partial f/\partial z$.

- Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$ (pre-math M5). It is needed for Section 9.1.2(ii).
- Newton's second law for a small piece of string (Plus Two mechanics).

1. What a wave is

A wave is a disturbance that moves through a medium with its shape fixed and its speed constant. The book adds that real waves may spread, fade or change shape, but the simplest case is a fixed shape moving at constant speed in one dimension (Fig. 9.1).

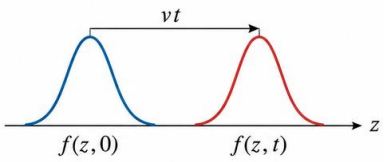


Fig. 9.1 (book): a wave of fixed shape at t = 0 and at a later time, shifted to the right by vt.

Call the shape at $t = 0$ $g(z) = f(z, 0)$. At a later time t, each point has moved a distance vt to the right. So the displacement at z at time t equals the displacement at $z - vt$ at time 0:

$$f(z, t) = f(z - vt, 0) = g(z - vt) \tag{9.1}$$

A function of z and t that depends only on the combination $z - vt$ describes a wave of fixed shape moving to the right at speed v. The book's examples f_1, f_2, f_3 are waves, because each depends on $z - vt$ alone. The examples f_4 and f_5 are not, because they do not.

2. Where the wave equation comes from

Take a taut string under tension T, with mass μ per unit length. A short piece between z and $z + \Delta z$ feels a net transverse force from the tension. For small displacements the angles are small, so $\sin\theta \approx \tan\theta = \partial f/\partial z$. The net force is

$$\Delta F = T\sin\theta' - T\sin\theta \approx T\left(\left.\frac{\partial f}{\partial z}\right|_{z+\Delta z} - \left.\frac{\partial f}{\partial z}\right|_{z}\right) \approx T\frac{\partial^2 f}{\partial z^2}\Delta z$$

Newton's second law for that piece gives $\Delta F = \mu\,\Delta z\,\partial^2 f/\partial t^2$. Equating the two and rearranging:

$$\frac{\partial^2 f}{\partial z^2} = \frac{1}{v^2}\frac{\partial^2 f}{\partial t^2} \tag{9.2}$$

with the speed of propagation

$$v = \sqrt{\frac{T}{\mu}} \tag{9.3}$$

Equation 9.2 is the **wave equation**. Its solutions are the functions of $z - vt$.

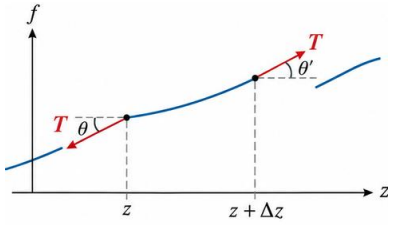


Fig. 9.2 (book): a string segment between z and z + Δz, with tension T at angles θ and θ′.

3. Solutions of the wave equation

Let $u = z - vt$ and $f = g(u)$. The chain rule gives

$$\frac{\partial f}{\partial z} = \frac{dg}{du}$$
$$\frac{\partial f}{\partial t} = -v\frac{dg}{du}$$
$$\frac{\partial^2 f}{\partial z^2} = \frac{d^2 g}{du^2}$$
$$\frac{\partial^2 f}{\partial t^2} = v^2\frac{d^2 g}{du^2}$$

Substituting into Eq. 9.2 shows that $f = g(z - vt)$ satisfies it for any differentiable g. The same argument shows that $h(z + vt)$ satisfies it too:

$$f(z, t) = g(z - vt) \tag{9.4}$$

$$f(z, t) = h(z + vt) \tag{9.5}$$

The wave equation is linear, so the sum of two solutions is also a solution. The most general solution is therefore a wave to the right plus a wave to the left:

$$f(z, t) = g(z - vt) + h(z + vt) \tag{9.6}$$

The book notes that every solution has this form. The class of solutions is much larger than for an ordinary second-order equation, because the wave equation is a partial differential equation.

4. Sinusoidal waves

(i) **Terminology.** The most familiar wave is the cosine:

$$f(z, t) = A\cos[k(z - vt) + \delta] \tag{9.7}$$

A is the amplitude, the maximum displacement. The argument of the cosine is the phase; δ is the phase constant. The wave is delayed by a distance δ/k. The wave number k gives the wavelength:

$$\lambda = \frac{2\pi}{k} \tag{9.8}$$

The period, the time for one complete oscillation at a fixed point, is

$$T = \frac{2\pi}{kv} \tag{9.9}$$

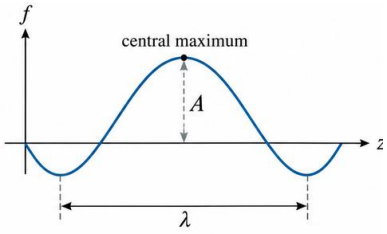


Fig. 9.3 (book): a sinusoidal wave at t = 0, with amplitude A, wavelength λ and the central maximum at δ/k.

PLUS TWO RECALL: FREQUENCY, WAVELENGTH AND SPEED

Recall. From Class 12: for a wave, speed = frequency × wavelength.

Re-derive.

1. In one period T the wave moves forward one wavelength λ.
2. So $v = \lambda/T$, and the frequency is $\nu = 1/T$.
3. Substituting T from Eq. 9.9 gives $\nu = kv/2\pi$.

Saved. $\nu = \frac{kv}{2\pi} = \frac{v}{\lambda} = \frac{1}{T}$ Eq. 9.10.

The frequency is the number of oscillations per second:

$$\nu = \frac{kv}{2\pi} = \frac{v}{\lambda} = \frac{1}{T} \tag{9.10}$$

It is more convenient to use the angular frequency $\omega = 2\pi\nu$. It is the number of radians swept per second, by analogy with circular motion:

$$\omega = 2\pi\nu = kv \tag{9.11}$$

Written in terms of ω, the wave is

$$f(z,t) = A\cos(kz - \omega t + \delta) \tag{9.12}$$

A wave going to the left is

$$f(z,t) = A\cos(kz + \omega t - \delta) \tag{9.13}$$

Because the cosine is even, this can also be written $A\cos(-kz - \omega t + \delta)$ Eq. 9.14. Reversing the sign of k therefore reverses the direction of travel and nothing else.

(ii) **Complex notation.** Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$ Eq. 9.15 lets the wave be written as the real part of an exponential:

$$f(z,t) = \mathrm{Re}\left[Ae^{i(kz-\omega t+\delta)}\right] \tag{9.16}$$

Define the complex wave function and the complex amplitude $\tilde{A} = Ae^{i\delta}$:

$$\tilde{f}(z,t) = \tilde{A}e^{i(kz-\omega t)} \tag{9.17}$$

$$f(z,t) = \mathrm{Re}[\tilde{f}(z,t)] \tag{9.18}$$

Example 9.1. Two waves with the same k and ω add by adding their complex amplitudes, $\tilde{A}_3 = \tilde{A}_1 + \tilde{A}_2$, which gives

$$A_3 e^{i\delta_3} = A_1 e^{i\delta_1} + A_2 e^{i\delta_2} \tag{9.19}$$

The real part of the sum is the sum of the real parts, so this works. It does not work for products: $\mathrm{Re}(\tilde{f}_1\tilde{f}_2) \neq \mathrm{Re}(\tilde{f}_1)\,\mathrm{Re}(\tilde{f}_2)$. Quadratic quantities such as energy have to be handled with the real fields.

(iii) **Linear combinations.** Any wave can be written as a sum of sinusoidal waves over all k (the book's Fourier representation):

$$\tilde{f}(z,t) = \int_{-\infty}^{\infty} \tilde{A}(k)e^{i(kz-\omega t)}\,dk \tag{9.20}$$

The coefficient $\tilde{A}(k)$ comes from the initial shape and velocity (Prob. 9.34). Here k runs over negative values as well, to include both directions. The book says that from here on only sinusoidal waves are needed: if you know how one sinusoidal wave behaves, you know how any wave behaves.

Problems 9.3 and 9.4 are listed in the book sheets but are not worked in these notes.

9.1.3 Reflection and transmission on a string

*Source: book-sources/m3-sec2-bc.html, Griffiths pp. 396–399. Equations 9.21–9.33 in full. This section is **background** (beyond the syllabus list, which takes items 9–14 from §9.1.1, §9.1.2 and §9.1.4). Problems 9.5–9.7 are not worked.*

Before you start

- Sinusoidal waves and complex amplitudes (Section 9.1.2).
- Continuity of a function and of its slope at a point (Beyond in Plus Two; pre-math M7).

1. Why the boundary matters

So far the string has been infinitely long. What a wave does at a join depends on how the string is attached. Tie two strings with the same tension T but different mass per unit length μ_1 and μ_2. Then the speeds $v_1 = \sqrt{T/\mu_1}$ and $v_2 = \sqrt{T/\mu_2}$ are different. Put the knot at $z = 0$.

2. The three waves

A sinusoidal wave of frequency ω comes in from the left on string 1. It gives a reflected wave going back along string 1, and a transmitted wave going on along string 2:

$$\tilde{f}_I = \tilde{A}_I e^{i(k_1 z - \omega t)} \quad (z < 0) \tag{9.21}$$

$$\tilde{f}_R = \tilde{A}_R e^{i(-k_1 z - \omega t)} \quad (z < 0) \tag{9.22}$$

$$\tilde{f}_T = \tilde{A}_T e^{i(k_2 z - \omega t)} \quad (z > 0) \tag{9.23}$$

The reflected wave has $-k_1$ because it travels to the left. All three have the same ω, which is set by the source. Because the speeds differ, the wave numbers differ:

$$\frac{\lambda_1}{\lambda_2} = \frac{k_2}{k_1} = \frac{v_1}{v_2} \tag{9.24}$$

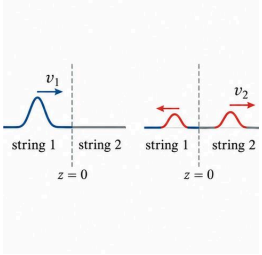


Fig. 9.5 (book): a pulse incident on a knot at z = 0, with the reflected and transmitted pulses.

A single sinusoid is an idealisation: a finite pulse is made of sinusoids of many frequencies (Eq. 9.20). The book uses sinusoids of infinite extent because only then is there a single frequency.

For $z < 0$ the total displacement is $\tilde{f}_I + \tilde{f}_R$, and for $z > 0$ it is $\tilde{f}_T$:

$$\tilde{f}(z,t) = \begin{cases} \tilde{A}_I e^{i(k_1 z - \omega t)} + \tilde{A}_R e^{i(-k_1 z - \omega t)}, & z < 0 \\ \tilde{A}_T e^{i(k_2 z - \omega t)}, & z > 0 \end{cases} \tag{9.25}$$

3. Boundary conditions

There must be no break in the string at the knot, so the displacement is continuous:

$$f(0^-, t) = f(0^+, t) \tag{9.26}$$

If the knot has negligible mass, a net force on it would give it an infinite acceleration. So the slope is continuous too (Fig. 9.6):

$$\left.\frac{\partial f}{\partial z}\right|_{0^-} = \left.\frac{\partial f}{\partial z}\right|_{0^+} \tag{9.27}$$

Because the imaginary part of $\tilde{f}$ differs from the real part only in cosine replaced by sine, the complex function obeys the same rules:

$$\begin{aligned} \tilde{f}(0^-, t) &= \tilde{f}(0^+, t) \\ \left.\frac{\partial \tilde{f}}{\partial z}\right|_{0^-} &= \left.\frac{\partial \tilde{f}}{\partial z}\right|_{0^+} \end{aligned} \tag{9.28}$$

4. Solving for the amplitudes

Apply Eq. 9.28 to Eq. 9.25. Continuity of $\tilde{f}$ gives $\tilde{A}_I + \tilde{A}_R = \tilde{A}_T$. Continuity of the slope gives $ik_1(\tilde{A}_I - \tilde{A}_R) = ik_2\tilde{A}_T$, so $k_1(\tilde{A}_I - \tilde{A}_R) = k_2\tilde{A}_T$. Solving the two:

$$\begin{aligned} \tilde{A}_R &= \frac{k_1 - k_2}{k_1 + k_2}\tilde{A}_I \\ \tilde{A}_T &= \frac{2k_1}{k_1 + k_2}\tilde{A}_I \end{aligned} \tag{9.29}$$

Using $k \propto 1/v$ (Eq. 9.24) gives the same result in terms of speeds:

$$\begin{aligned} \tilde{A}_R &= \frac{v_2 - v_1}{v_2 + v_1}\tilde{A}_I \\ \tilde{A}_T &= \frac{2v_2}{v_2 + v_1}\tilde{A}_I \end{aligned} \tag{9.30}$$

Real amplitudes and phases:

$$A_R e^{i\delta_R} = \frac{v_2 - v_1}{v_2 + v_1} A_I e^{i\delta_I}$$
$$A_T e^{i\delta_T} = \frac{2v_2}{v_2 + v_1} A_I e^{i\delta_I} \qquad (9.31)$$

If the second string is lighter ($v_2 > v_1$), all three waves are in phase, and

$$A_R = \frac{v_2 - v_1}{v_2 + v_1} A_I, \qquad A_T = \frac{2v_2}{v_2 + v_1} A_I \quad (9.32)$$

If the second string is heavier ($v_2 < v_1$), the reflected wave is turned upside down: its phase is shifted by π. Since $\cos(\theta - \pi) = -\cos\theta$, the amplitudes are

$$A_R = \frac{v_1 - v_2}{v_2 + v_1} A_I, \qquad A_T = \frac{2v_2}{v_2 + v_1} A_I \quad (9.33)$$

In the limit where the second string is fixed to an immovable post, the transmitted amplitude is zero and the reflected amplitude equals the incident one: all of the wave comes back.

9.1.4 Polarization of transverse waves

Source: book-sources/m3-sec2-polar.html (and the first lines of the sheet on p. 401), Griffiths pp. 399–401. Equations 9.34–9.39 in full. Problem 9.8 (circular polarization) is described in the text but not worked.

Before you start

- Vectors and the dot product (pre-math M9, M10).
- Sinusoidal waves (Section 9.1.2).

1. Transverse and longitudinal waves

On a string, shaking it up and down gives a **transverse** wave: the displacement is perpendicular to the direction of travel. A slinky can also carry a **longitudinal** (compression) wave, where the displacement is along the direction of travel. Sound in air is longitudinal. Electromagnetic waves are transverse, as Section 9.2 shows.

2. Two independent transverse directions

Perpendicular to the direction of travel there are two independent directions. A transverse wave can therefore be polarized in two independent ways. Shaking the string up and down (vertical) gives

$$\tilde{\mathbf{f}}^v = \tilde{A} e^{i(kz - \omega t)} \hat{x} \qquad (9.34)$$

and shaking it left and right (horizontal) gives

$$\tilde{\mathbf{f}}^h = \tilde{A} e^{i(kz - \omega t)} \hat{y} \qquad (9.35)$$

Any other direction in the xy plane is a combination:

$$\tilde{\mathbf{f}} = \tilde{A} e^{i(kz - \omega t)} \hat{n} \qquad (9.36)$$

$\hat{n}$ is the **polarization vector**. It defines the plane in which the string vibrates (Fig. 9.8).

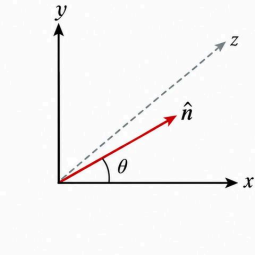


Fig. 9.8c (book): a wave polarized along n̂, at angle θ to the x axis in the xy plane.

For a transverse wave, $\hat{n}$ is perpendicular to the direction of travel:

$$\hat{n} \cdot \hat{z} = 0 \qquad (9.37)$$

If θ is the angle that $\hat{n}$ makes with the x axis in the xy plane:

$$\hat{n} = \cos\theta\,\hat{x} + \sin\theta\,\hat{y} \tag{9.38}$$

Then the wave of Fig. 9.8c is the sum of a horizontal and a vertical wave with amplitudes $\tilde{A}\cos\theta$ and $\tilde{A}\sin\theta$:

$$\begin{aligned}\tilde{\mathbf{f}} = {} & (\tilde{A}\cos\theta)e^{i(kz-\omega t)}\hat{x} \\ & + (\tilde{A}\sin\theta)e^{i(kz-\omega t)}\hat{y}\end{aligned} \tag{9.39}$$

3. Notes on the sign and on circular polarization

Changing the sign of $\hat{n}$ is equivalent to advancing the phase constant by $180°$, since both change the sign of the wave (footnote 4). The book uses the convention that $\hat{n}$ and the phase are chosen together.

If the two components in Eq. 9.39 have equal amplitude but are $90°$ out of phase, the string moves in a circle. That is circular polarization. Problem 9.8 asks which way the circle turns, and how to produce it; the book's optics terms are right circular and left circular.

The notation for elliptical or circular polarization is simplified in optics by using a complex $\hat{n}$. The book does not use it.

Section 3: electromagnetic waves in vacuum (Griffiths 9.2)

9.2.1 The wave equation for E and B

Source: book-sources/m3-sec3-wave-eq.html and the end of m3-sec2-polar.html, Griffiths pp. 401–402. Equations 9.40–9.43 in full.

Before you start

- Maxwell's equations in vacuum with no charge or current (Module 2, Section 7.3).
- Curl of a curl: $\nabla\times(\nabla\times\mathbf{A}) = \nabla(\nabla\cdot\mathbf{A}) - \nabla^2\mathbf{A}$ (Beyond in Plus Two; pre-math M13).

1. Maxwell's equations in empty space

With no charge and no current, Maxwell's equations are four first-order equations:

$$\begin{aligned}&\text{(i)}\ \nabla\cdot\mathbf{E} = 0 \\ &\text{(iii)}\ \nabla\times\mathbf{E} = -\frac{\partial\mathbf{B}}{\partial t} \\ &\text{(ii)}\ \nabla\cdot\mathbf{B} = 0 \\ &\text{(iv)}\ \nabla\times\mathbf{B} = \mu_0\epsilon_0\frac{\partial\mathbf{E}}{\partial t}\end{aligned} \tag{9.40}$$

These are coupled. $\mathbf{E}$ and $\mathbf{B}$ appear in each other's equations. The book decouples them.

2. Decoupling

Take the curl of (iii) and of (iv). For (iii):

$$\begin{aligned}\nabla \times (\nabla \times \mathbf{E}) &= \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} \\ &= -\frac{\partial}{\partial t}(\nabla \times \mathbf{B}) \\ &= -\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}\end{aligned}$$

The last step uses (iv). Then (i) removes the divergence term, and the same method applied to (iv) gives the equation for $\mathbf{B}$:

$$\begin{aligned}\nabla^2 \mathbf{E} &= \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} \\ \nabla^2 \mathbf{B} &= \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2}\end{aligned} \qquad (9.41)$$

Each equation is second order and decoupled, but the price is that the equations are second order. Each Cartesian component of $\mathbf{E}$ and $\mathbf{B}$ satisfies the three-dimensional wave equation $\nabla^2 f = \frac{1}{v^2}\partial^2 f/\partial t^2$, which generalises Eq. 9.2 with $\partial^2 f/\partial z^2$ replaced by $\nabla^2 f$.

PLUS TWO RECALL: SPEED OF LIGHT FROM ϵ_0 AND μ_0

Recall. From Class 12: electromagnetic waves travel in vacuum at $c = 1/\sqrt{\epsilon_0 \mu_0}$, the same as the speed of light.

Re-derive.

1. Start from the wave equation Eq. 9.41 for $\mathbf{E}$: $\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \, \partial^2 \mathbf{E}/\partial t^2$.
2. For a wave $f(z - vt)$, the same equation with $\partial^2 f/\partial z^2 = (1/v^2)\partial^2 f/\partial t^2$ (Eq. 9.2) gives $1/v^2 = \mu_0 \epsilon_0$.
3. Therefore $v = 1/\sqrt{\mu_0 \epsilon_0}$.

Saved. $c = 1/\sqrt{\epsilon_0 \mu_0}$ Eq. 9.43.

3. The speed

Comparing Eq. 9.41 with the wave equation gives the speed of the waves in vacuum:

RESULT 9.5: SPEED OF ELECTROMAGNETIC WAVES IN VACUUM

$$v = \frac{1}{\sqrt{\epsilon_0 \mu_0}} = 3.00 \times 10^8 \ \mathrm{m/s} \qquad (9.42)$$

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}} \qquad (9.43)$$

This is the measured speed of light, so light is an electromagnetic wave (footnote 6).

4. Why this matters

Maxwell's contribution to Ampère's law, the term $\mu_0 \epsilon_0 \, \partial \mathbf{E}/\partial t$, is what produces the wave equation. Without it, the equations would not admit waves. The constants ϵ_0 and μ_0 are measured with charges, batteries and wires, and yet they give c. The book calls this the crucial role of displacement current in the theory of light.

9.2.2 Monochromatic plane waves

Source: book-sources/m3-sec3-plane.html (and the first sheet, m3-sec3-wave-eq.html), Griffiths pp. 402–406. Equations 9.44–9.53, Table 9.1 and Example 9.2 in full. Figure 9.9–9.11 are redrawn. Problem 9.9 is not worked.

Before you start

- Complex exponentials and the real part (Section 9.1.2, pre-math M5, M6).
- Faraday's law in differential form (Eq. 9.40, iii).
- The dot product and the cross product (pre-math M8, M9, M10).

1. Monochromatic and plane

A **monochromatic** wave has one frequency ω; its colour is set by ω in the visible range. A **plane** wave is uniform over every plane perpendicular to the direction of travel (Fig. 9.9). Take the direction of travel to be z, with no x or y dependence. Then the fields have the form

$$\begin{aligned}\tilde{\mathbf{E}}(z,t) &= \tilde{\mathbf{E}}_0 e^{i(kz-\omega t)} \\ \tilde{\mathbf{B}}(z,t) &= \tilde{\mathbf{B}}_0 e^{i(kz-\omega t)}\end{aligned} \qquad (9.44)$$

with $\omega = ck$. The complex amplitudes $\tilde{\mathbf{E}}_0$, $\tilde{\mathbf{B}}_0$ carry direction and phase; the physical fields are the real parts.

Table 9.1 gives the spectrum. Radio waves have wavelengths of metres and more; visible light has wavelengths from about 3.0×10^{-7} m (near ultraviolet) to 1.0×10^{-6} m (near infrared). The table is reproduced below.

Frequency (Hz)	Type	Wavelength (m)
10^{22}		10^{-13}
10^{21}	Gamma rays	10^{-12}
10^{20}		10^{-11}
10^{19}		10^{-10}
10^{18}	X rays	10^{-9}
10^{17}		10^{-8}
10^{16}	Ultraviolet	10^{-7}
10^{15}	Visible	10^{-6}
10^{14}	Infrared	10^{-5}
10^{13}		10^{-4}
10^{12}		10^{-3}
10^{11}		10^{-2}
10^{10}	Microwave	10^{-1}
10^{9}		1
10^{8}	TV, FM	10
10^{7}		10^{2}
10^{6}	AM	10^{3}
10^{5}		10^{4}
10^{4}	RF	10^{5}
10^{3}		10^{6}

2. Constraints from Maxwell's equations

Plane waves of the form Eq. 9.44 satisfy the wave equation, but Maxwell's equations impose more. Because $\nabla \cdot \mathbf{E} = 0$ and $\nabla \cdot \mathbf{B} = 0$ and the fields do not depend on x or y:

$$(\tilde{E}_0)_z = (\tilde{B}_0)_z = 0 \qquad (9.45)$$

So $\mathbf{E}$ and $\mathbf{B}$ are both perpendicular to the direction of travel. Electromagnetic waves are **transverse**. (Footnote 8: the real part and the imaginary part obey the same equations, so the complex solution is valid.)

Faraday's law fixes the relation between the two amplitudes:

$$\begin{aligned}-k(\tilde{E}_0)_y &= \omega(\tilde{B}_0)_x \\ k(\tilde{E}_0)_x &= \omega(\tilde{B}_0)_y\end{aligned} \qquad (9.46)$$

which is the same as

$$\tilde{\mathbf{B}}_0 = \frac{k}{\omega}(\hat{z} \times \tilde{\mathbf{E}}_0) \qquad (9.47)$$

So $\mathbf{E}$ and $\mathbf{B}$ are perpendicular to each other and to $\hat{z}$. They are in phase, and their real amplitudes are related by

$$B_0 = \frac{k}{\omega}E_0 = \frac{1}{c}E_0 \qquad (9.48)$$

PLUS TWO RECALL: TRANSVERSE, IN-PHASE FIELDS

Recall. From Class 12: in an electromagnetic wave, $\mathbf{E}$ and $\mathbf{B}$ oscillate in phase, at right angles to each other and to the direction of travel.

Re-derive.

1. Assume $\tilde{\mathbf{E}}$ has only an x component. Then $\mathbf{B}$ must be in the y direction, by Eq. 9.47 with $\hat{z} \times \hat{x} = \hat{y}$.
2. Faraday's law: $\nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t$ gives $kE_0 = \omega B_0$.
3. Both fields have the same phase $kz - \omega t$, so they vanish and peak together.

Saved. $\mathbf{E} \perp \mathbf{B} \perp \hat{z}$, in phase, with $B_0 = E_0/c$ Eq. 9.48.

3. Example 9.2

Take $\mathbf{E}$ along x. Then $\mathbf{B}$ is along y by Eq. 9.47, and

$$\tilde{\mathbf{E}} = \tilde{E}_0 e^{i(kz-\omega t)}\hat{x}$$
$$\tilde{\mathbf{B}} = \frac{1}{c}\tilde{E}_0 e^{i(kz-\omega t)}\hat{y}$$

Taking the real parts gives the physical fields:

$$\mathbf{E}(z,t) = E_0 \cos(kz - \omega t + \delta)\hat{x},$$
$$\mathbf{B}(z,t) = \frac{1}{c}E_0 \cos(kz - \omega t + \delta)\hat{y} \qquad (9.49)$$

This is the model for a monochromatic plane wave (Fig. 9.10). The wave is said to be polarized in the x direction, by convention the direction of $\mathbf{E}$.

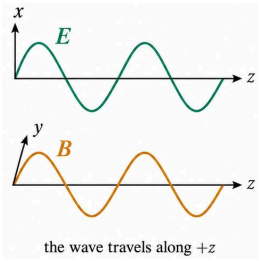


Fig. 9.10 (book): monochromatic plane wave with E along x and B along y, traveling along z.

4. Arbitrary direction

For a wave travelling along a unit vector $\hat{k}$, define the propagation vector $\mathbf{k} = k\hat{k}$. The phase kz is replaced by $\mathbf{k} \cdot \mathbf{r}$ (Fig. 9.11):

$$\tilde{\mathbf{E}}(\mathbf{r},t) = \tilde{E}_0 e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}\hat{n}$$
$$\tilde{\mathbf{B}}(\mathbf{r},t) = \frac{1}{c}\tilde{E}_0 e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}(\hat{k}\times\hat{n}) = \frac{1}{c}\hat{k}\times\tilde{\mathbf{E}} \qquad (9.50)$$

$\hat{n}$ is the polarization vector, and transversality is $\hat{n}\cdot\hat{k} = 0$ Eq. 9.51. The real fields are

$$\mathbf{E}(\mathbf{r},t) = E_0 \cos(\mathbf{k}\cdot\mathbf{r} - \omega t + \delta)\hat{n} \qquad (9.52)$$

$$\mathbf{B}(\mathbf{r},t) = \frac{1}{c}E_0 \cos(\mathbf{k}\cdot\mathbf{r} - \omega t + \delta)(\hat{k}\times\hat{n}) \qquad (9.53)$$

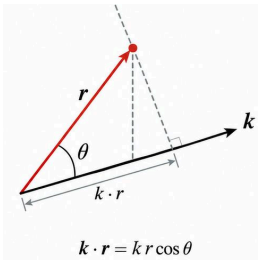


Fig. 9.11 (book): the propagation vector k and the position vector r; the phase is k · r.

Problem 9.9 (fields for waves in other directions) is outside these notes.

9.2.3 Energy and momentum in electromagnetic waves

Source: book-sources/m3-sec3-energy.html, Griffiths pp. 406–408. Equations 9.54–9.65 in full, and footnotes 9–12. Problems 9.10–9.13 are described; Problems 9.11 and 9.12 are not worked. The Problem 9.13 sheet also carries the §9.3 heading (see m3-sec4-linear.html).

Before you start

- The energy density u and the Poynting vector $\mathbf{S}$ (Section 8.1.2, Eqs. 8.5 and 8.10).
- Average of $\cos^2$ over a cycle (pre-math M4).
- Momentum and force per unit area (Plus Two mechanics).

1. Energy density of a plane wave

From Eq. 8.5, the energy per unit volume in the fields is

$$u = \frac{1}{2}\left(\epsilon_0 E^2 + \frac{1}{\mu_0}B^2\right) \tag{9.54}$$

For a monochromatic plane wave (Eq. 9.49 or 9.53), $B = E/c$, so $B^2/\mu_0 = \epsilon_0 E^2$ (since $1/(\mu_0 c^2) = \epsilon_0$). Then

$$B^2 = \frac{1}{c^2}E^2 = \mu_0\epsilon_0 E^2 \tag{9.55}$$

and the electric and magnetic parts of the energy are equal:

$$u = \epsilon_0 E^2 = \epsilon_0 E_0^2 \cos^2(kz - \omega t + \delta) \tag{9.56}$$

> **PLUS TWO RECALL: ENERGY DENSITY OF AN ELECTRIC FIELD**
>
> **Recall.** From Class 12: a field E in vacuum stores energy $\frac{1}{2}\epsilon_0 E^2$ per unit volume.
>
> **Re-derive.**
>
> 1. Use the magnetic result Eq. 8.5, $\frac{1}{2\mu_0}B^2$, for the second term.
> 2. With $B = E/c$ Eq. 9.48 and $c^2 = 1/(\epsilon_0\mu_0)$, $\frac{1}{2\mu_0}B^2 = \frac{1}{2}\epsilon_0 E^2$.
> 3. So the magnetic term equals the electric term, and each is $\frac{1}{2}\epsilon_0 E^2$.
>
> **Saved.** $u = \epsilon_0 E^2$ Eq. 9.56, with the electric and magnetic parts equal.

2. Energy flux

The Poynting vector gives the energy flux density:

$$\mathbf{S} = \frac{1}{\mu_0}(\mathbf{E}\times\mathbf{B}) \tag{9.57}$$

For a wave travelling along z, $\mathbf{E}\times\mathbf{B}$ points along z, and

$$\mathbf{S} = c\epsilon_0 E_0^2 \cos^2(kz - \omega t + \delta)\hat{z} = cu\,\hat{z} \tag{9.58}$$

So $\mathbf{S}$ equals the energy density times the velocity of the wave. Check: in time Δt the energy in a cylinder of cross section A and length $c\,\Delta t$ passes through area A, which is $uAc\Delta t$. Dividing by $A\Delta t$ gives uc, as Eq. 9.58 says.

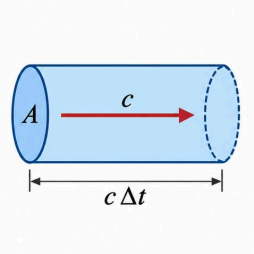


Fig. 9.12 (book): a cylinder of cross section A and length cΔt; the energy in it crosses A in time Δt.

3. Momentum density

Fields carry momentum as well as energy. Eq. 8.29 gives the momentum density:

$$\mathbf{g} = \frac{1}{c^2}\mathbf{S} \tag{9.59}$$

For a plane wave, $\mathbf{g} = (u/c)\hat{z}$ (Eq. 9.60). Footnote 9 warns against taking the real part of a complex energy density: $\mathrm{Re}(\tilde{u})$ gives $\epsilon_0 E_0^2\cos[2(\mathbf{k}\cdot\mathbf{r} - \omega t + \delta)]$, which is not u. Use the real fields.

$$\begin{aligned}\mathbf{g} &= \frac{1}{c}\epsilon_0 E_0^2 \cos^2(kz - \omega t + \delta)\hat{z} \\ &= \frac{1}{c}u\,\hat{z}\end{aligned} \tag{9.60}$$

4. Time averages and intensity

Light has a wavelength near 5×10^{-7} m and a period near 10^{-15} s. Any measurement covers many cycles, so only the average over a cycle matters. Since the average of $\cos^2$ over one period is $\frac{1}{2}$ (footnote 10), the averages are

$$\langle u \rangle = \frac{1}{2}\epsilon_0 E_0^2 \tag{9.61}$$

$$\langle \mathbf{S} \rangle = \frac{1}{2} c \epsilon_0 E_0^2 \hat{z} \tag{9.62}$$

$$\langle \mathbf{g} \rangle = \frac{1}{2c}\epsilon_0 E_0^2 \hat{z} \tag{9.63}$$

The average power per unit area is the **intensity**:

$$I \equiv \langle S \rangle = \frac{1}{2} c \epsilon_0 E_0^2 \tag{9.64}$$

5. Radiation pressure

Light at normal incidence on a perfect absorber delivers momentum to the surface. In time Δt the momentum transferred to area A is $\Delta p = \langle g \rangle A c \Delta t$. The radiation pressure is the force per unit area:

$$\begin{aligned} P &= \frac{1}{A}\frac{\Delta p}{\Delta t} \\ &= \frac{1}{2}\epsilon_0 E_0^2 \\ &= \frac{I}{c} \end{aligned} \tag{9.65}$$

A perfect reflector gives twice this value, because the momentum reverses direction. The naïve account in terms of the electric field driving charges, which then feel $q\mathbf{v} \times \mathbf{B}$, is qualitative. Problem 9.11 shows that it is not complete: the velocity is 90° out of phase with the fields, so there must be a damping force to absorb energy.

Problem 9.10 asks for the radiation pressure of sunlight of intensity about 1300 W/m^2 on an absorber and on a reflector. Problems 9.11–9.13 (magnetic force on a particle, complex averages, the Maxwell stress tensor) are listed in the book but are not worked here.

Section 4: electromagnetic waves in matter (Griffiths 9.3)

9.3.1–9.3.2 Propagation in linear media; reflection and transmission at normal incidence

Source: book-sources/m3-sec4-linear.html (and the first lines of m3-sec4-oblique.html for Eqs. 9.87–9.89), Griffiths pp. 408–412. Equations 9.66–9.89 in full, footnotes 13–16 and Problems 9.14–9.15. §9.3.1 is in the syllabus (item 14); §9.3.2 is background.

Before you start

- Maxwell's equations in vacuum (Eq. 9.40) and the plane-wave relations Eq. 9.46–Eq. 9.48.
- The boundary conditions between two media, Eq. 7.74 (Module 2, Section 7.3.6), given here as Eq. 9.75.
- Permittivity and permeability of a medium, $\epsilon = \epsilon_r \epsilon_0$ (pre-physics P12, P13).

1. Maxwell's equations in a linear medium

Inside matter, with no free charge or free current, Maxwell's equations are Eq. 9.66:

$$\begin{aligned} \text{(i)}\ \nabla \cdot \mathbf{D} &= 0 \\ \text{(iii)}\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\ \text{(ii)}\ \nabla \cdot \mathbf{B} &= 0 \\ \text{(iv)}\ \nabla \times \mathbf{H} &= \frac{\partial \mathbf{D}}{\partial t} \end{aligned} \qquad (9.66)$$

For a linear medium, $\mathbf{D} = \epsilon \mathbf{E}$ and $\mathbf{H} = \mathbf{B}/\mu$ Eq. 9.67. If the medium is also homogeneous, so ϵ and μ are constant, the equations take the vacuum form with $\mu_0\epsilon_0$ replaced by $\mu\epsilon$ Eq. 9.68:

$$\begin{aligned} \text{(i)}\ \nabla \cdot \mathbf{E} &= 0 \\ \text{(iii)}\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\ \text{(ii)}\ \nabla \cdot \mathbf{B} &= 0 \\ \text{(iv)}\ \nabla \times \mathbf{B} &= \mu\epsilon \frac{\partial \mathbf{E}}{\partial t} \end{aligned} \qquad (9.68)$$

2. Speed and index of refraction

The wave equation in the medium has the same form as in vacuum, with $\mu_0\epsilon_0 \to \mu\epsilon$. So the speed in the medium is

PLUS TWO RECALL: REFRACTIVE INDEX AS RATIO OF SPEEDS

Recall. From Class 12: the refractive index of a medium is $n = c/v$, the ratio of the speed of light in vacuum to its speed in the medium.

Re-derive.

1. From Eq. 9.68 the wave equation has $\mu\epsilon$ in place of $\mu_0\epsilon_0$.
2. Comparing with the vacuum wave equation gives $v^2 = 1/\mu\epsilon$, so $v = 1/\sqrt{\mu\epsilon}$.
3. Write $v = c/n$; then $n = c/v = \sqrt{\mu\epsilon/\mu_0\epsilon_0}$.

Saved. $v = c/n$ Eq. 9.69 with n as in Eq. 9.70.

$$\begin{aligned} v &= \frac{1}{\sqrt{\epsilon\mu}} = \frac{c}{n} \\ n &\equiv \sqrt{\frac{\epsilon\mu}{\epsilon_0\mu_0}} \end{aligned} \qquad (9.69)$$

n is the **index of refraction** Eq. 9.70. Most materials have $\mu \approx \mu_0$, so

$$n \approx \sqrt{\epsilon_r} \qquad (9.71)$$

where ϵ_r is the dielectric constant (Eq. 4.34). Since $\epsilon_r > 1$ for almost all materials, light travels more slowly in matter.

Footnote 13 (book): the field inside a material is the sum of the original field and the fields of the molecular dipoles it induces. Together they make a single wave with the same frequency and a different speed. This "Ewald–Oseen extinction theorem" explains transparency. It depends on linearity.

Footnote 14 (book): ϵ_r is independent of the amplitude of $\mathbf{E}$, but it may depend on frequency. The static dielectric constant of water gives $n = 8.9$, which is far from the visible-light value $n \approx 1.33$.

3. Energy and the Poynting vector in a medium

All earlier results carry over with $\epsilon_0 \to \epsilon$, $\mu_0 \to \mu$, $c \to v$. The energy density (footnote 15: see Prob. 8.27) and the Poynting vector are

$$u = \frac{1}{2}\left(\epsilon E^2 + \frac{1}{\mu}B^2\right) \tag{9.72}$$

$$\mathbf{S} = \frac{1}{\mu}(\mathbf{E} \times \mathbf{B}) \tag{9.73}$$

For a monochromatic plane wave, $\omega = kv$ Eq. 9.11; $B_0 = E_0/v$ (9.48 with $c \to v$); and the intensity, the average power per unit area, is

$$I = \frac{1}{2}\epsilon v E_0^2 \tag{9.74}$$

Footnote 16: the momentum carried by a wave in matter is a matter of current debate.

4. Boundary conditions between two media

Chapter 7 gives the boundary conditions at an interface (Eq. 7.74). In the form used here Eq. 9.75: the normal component of ϵE and the normal component of B are continuous, and so are the tangential component of $\mathbf{E}$ and of $\mathbf{H} = \mathbf{B}/\mu$:

$$\begin{aligned} &\text{(i)}\ \epsilon_1 E_1^\perp = \epsilon_2 E_2^\perp \\ &\text{(iii)}\ \mathbf{E}_1^\parallel = \mathbf{E}_2^\parallel \\ &\text{(ii)}\ B_1^\perp = B_2^\perp \\ &\text{(iv)}\ \frac{1}{\mu_1}\mathbf{B}_1^\parallel = \frac{1}{\mu_2}\mathbf{B}_2^\parallel \end{aligned} \tag{9.75}$$

5. Normal incidence (§9.3.2)

Let the xy plane be the interface, with medium 1 on the left ($z < 0$) and medium 2 on the right. A plane wave of frequency ω, polarized in x, travels in the $+z$ direction and meets the interface head on (Fig. 9.13):

$$\begin{aligned} \tilde{\mathbf{E}}_I &= \tilde{E}_{0I} e^{i(k_1 z - \omega t)}\hat{x} \\ \tilde{\mathbf{B}}_I &= \frac{1}{v_1}\tilde{E}_{0I} e^{i(k_1 z - \omega t)}\hat{y} \end{aligned} \tag{9.76}$$

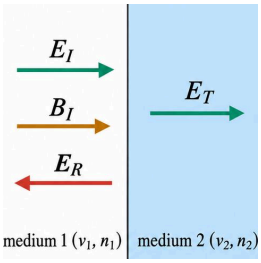


Fig. 9.13 (book): normal incidence on the interface between medium 1 and medium 2.

A reflected wave goes back into medium 1. Its $\mathbf{B}$ has the opposite sign, as Eq. 9.50 requires (the Poynting vector points along the direction of propagation):

$$\begin{aligned} \tilde{\mathbf{E}}_R &= \tilde{E}_{0R} e^{i(-k_1 z - \omega t)}\hat{x} \\ \tilde{\mathbf{B}}_R &= -\frac{1}{v_1}\tilde{E}_{0R} e^{i(-k_1 z - \omega t)}\hat{y} \end{aligned} \tag{9.77}$$

A transmitted wave goes on into medium 2:

$$\begin{aligned} \tilde{\mathbf{E}}_T &= \tilde{E}_{0T} e^{i(k_2 z - \omega t)}\hat{x} \\ \tilde{\mathbf{B}}_T &= \frac{1}{v_2}\tilde{E}_{0T} e^{i(k_2 z - \omega t)}\hat{y} \end{aligned} \tag{9.78}$$

At $z = 0$ there are no components perpendicular to the surface, so (i) and (ii) are trivial. Condition (iii) gives

$$\tilde{E}_{0I} + \tilde{E}_{0R} = \tilde{E}_{0T} \tag{9.79}$$

and (iv) gives

$$\frac{1}{\mu_1}\left(\frac{1}{v_1}\tilde{E}_{0I} - \frac{1}{v_1}\tilde{E}_{0R}\right) = \frac{1}{\mu_2}\frac{1}{v_2}\tilde{E}_{0T} \tag{9.80}$$

i.e. $\tilde{E}_{0I} - \tilde{E}_{0R} = \beta\tilde{E}_{0T}$ Eq. 9.81, with

$$\beta \equiv \frac{\mu_1 v_1}{\mu_2 v_2} = \frac{\mu_1 n_2}{\mu_2 n_1} \tag{9.82}$$

Solving Eq. 9.79 and Eq. 9.81:

$$\begin{aligned} \tilde{E}_{0R} &= \left(\frac{1-\beta}{1+\beta}\right)\tilde{E}_{0I} \\ \tilde{E}_{0T} &= \left(\frac{2}{1+\beta}\right)\tilde{E}_{0I} \end{aligned} \tag{9.83}$$

For $\mu \approx \mu_0$, $\beta = v_1/v_2$, and these become the string results Eq. 9.84. The real amplitudes are then Eq. 9.85, or in terms of indices Eq. 9.86:

$$\begin{aligned} \tilde{E}_{0R} &= \left(\frac{v_2 - v_1}{v_2 + v_1}\right)\tilde{E}_{0I} \\ \tilde{E}_{0T} &= \left(\frac{2v_2}{v_2 + v_1}\right)\tilde{E}_{0I} \end{aligned} \tag{9.84}$$

$$E_{0R} = \left|\frac{n_1 - n_2}{n_1 + n_2}\right| E_{0I}$$
$$E_{0T} = \left(\frac{2n_1}{n_1 + n_2}\right) E_{0I} \tag{9.86}$$

$$E_{0R} = \left|\frac{v_2 - v_1}{v_2 + v_1}\right| E_{0I}$$
$$E_{0T} = \left(\frac{2v_2}{v_2 + v_1}\right) E_{0I} \tag{9.85}$$

Eq. 9.85 gives the real amplitudes; the absolute value is needed because the sign of the reflected wave is carried by its phase. If $v_2 > v_1$ the reflected wave is right side up; if $v_2 < v_1$ it is upside down.

6. Reflection and transmission coefficients

The intensity is Eq. 9.74. With $\mu_1 = \mu_2 = \mu_0$, the reflection and transmission coefficients (fractions of incident intensity) are

$$R \equiv \frac{I_R}{I_I} = \left(\frac{E_{0R}}{E_{0I}}\right)^2 = \left(\frac{n_1 - n_2}{n_1 + n_2}\right)^2 \tag{9.87}$$

$$T \equiv \frac{I_T}{I_I} = \frac{\epsilon_2 v_2}{\epsilon_1 v_1}\left(\frac{E_{0T}}{E_{0I}}\right)^2 = \frac{4n_1 n_2}{(n_1 + n_2)^2} \tag{9.88}$$

Energy conservation requires

$$R + T = 1 \tag{9.89}$$

For air ($n_1 = 1$) to glass ($n_2 = 1.5$): $R = 0.04$ and $T = 0.96$. Most of the light is transmitted.

Problem 9.14: exact coefficients without assuming $\mu_1 = \mu_2 = \mu_0$. Problem 9.15: the reflected and transmitted waves keep the polarization of the incident wave.

9.3.3 Reflection and transmission at oblique incidence

*Source: book-sources/m3-sec4-oblique.html, Griffiths pp. 411–418. Equations 9.90–9.118 in full; footnotes 17–19. **Background** (beyond the syllabus list; item 14 needs §9.3.1 only). The laws of reflection and refraction are the part most likely to be useful in a question. Problems 9.16–9.19 are listed, not worked.*

Before you start

- Normal incidence and the interface conditions (Section 9.3.2, Eqs. 9.75–9.86).
- Snell's law $n_1 \sin\theta_1 = n_2 \sin\theta_2$ (pre-physics P5).
- Dot product of wave vectors and the exponent $\mathbf{k} \cdot \mathbf{r}$ (pre-math M10).

1. Setup

Now the wave meets the boundary at an arbitrary angle θ_I (Fig. 9.14). Normal incidence is the case $\theta_I = 0$. The incident, reflected and transmitted plane waves are

$$\begin{aligned} \tilde{\mathbf{E}}_I &= \tilde{\mathbf{E}}_{0I} e^{i(\mathbf{k}_I \cdot \mathbf{r} - \omega t)} \\ \tilde{\mathbf{B}}_I &= \frac{1}{v_1}(\hat{k}_I \times \tilde{\mathbf{E}}_I) \end{aligned} \tag{9.90}$$

$$\begin{aligned} \tilde{\mathbf{E}}_R &= \tilde{\mathbf{E}}_{0R} e^{i(\mathbf{k}_R \cdot \mathbf{r} - \omega t)} \\ \tilde{\mathbf{B}}_R &= \frac{1}{v_1}(\hat{k}_R \times \tilde{\mathbf{E}}_R) \end{aligned} \tag{9.91}$$

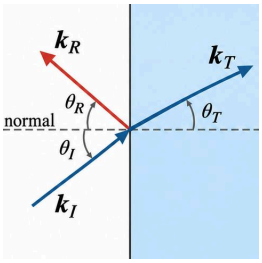


Fig. 9.14 (book): oblique incidence; wave vectors k_I, k_R, k_T and angles θ_I, θ_R, θ_T to the normal.

$$\begin{aligned} \tilde{\mathbf{E}}_T &= \tilde{\mathbf{E}}_{0T} e^{i(\mathbf{k}_T \cdot \mathbf{r} - \omega t)} \\ \tilde{\mathbf{B}}_T &= \frac{1}{v_2}(\hat{k}_T \times \tilde{\mathbf{E}}_T) \end{aligned} \tag{9.92}$$

All three have the same frequency, set by the source (footnote 17: non-linear media could change it). The wave numbers are related by

$$\begin{gathered} k_I v_1 = k_R v_1 = k_T v_2 = \omega \\ \text{so} \\ k_I = k_R = \frac{v_2}{v_1} k_T = \frac{n_1}{n_2} k_T \end{gathered} \tag{9.93}$$

2. Matching at the interface

The boundary conditions Eq. 9.75 must hold at every point of the interface and at every time. Each term has the form $e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}$, so the exponential factors must agree at $z = 0$:

$$\begin{gathered} \mathbf{k}_I \cdot \mathbf{r} = \mathbf{k}_R \cdot \mathbf{r} = \mathbf{k}_T \cdot \mathbf{r} \\ \text{when } z = 0 \end{gathered} \tag{9.95}$$

In components, for all x and y Eq. 9.96:

$$\begin{aligned} x(k_I)_x + y(k_I)_y &= x(k_R)_x + y(k_R)_y \\ &= x(k_T)_x + y(k_T)_y \end{aligned} \tag{9.96}$$

This is possible only if the coefficients of x and of y match separately, so $(k_I)_y = (k_R)_y = (k_T)_y$ Eq. 9.97 and $(k_I)_x = (k_R)_x = (k_T)_x$ Eq. 9.98. Orient the axes so that $\mathbf{k}_I$ lies in the xz plane; then so do $\mathbf{k}_R$ and $\mathbf{k}_T$. Time dependence matches automatically, which confirms the frequencies.

The exponentials cancel once Eq. 9.95 holds, and the matching conditions Eq. 9.94 become conditions on amplitudes.

3. The three laws of geometrical optics

First law. The incident, reflected and transmitted wave vectors lie in one plane, the **plane of incidence**, which also contains the normal to the surface.

Second law. From Eq. 9.98, $k_I \sin\theta_I = k_R \sin\theta_R = k_T \sin\theta_T$ Eq. 9.99. With Eq. 9.93, the angle of incidence equals the angle of reflection:

$$\theta_I = \theta_R \tag{9.100}$$

Third law (Snell's law). Using Eq. 9.93 in Eq. 9.99:

$$\frac{\sin\theta_T}{\sin\theta_I} = \frac{n_1}{n_2} \tag{9.101}$$

These laws use only the general form of the boundary conditions. The electrodynamics enters later, in the amplitudes. So any wave, including water or sound waves, obeys the same laws when it crosses from one medium into another.

4. Boundary conditions for the amplitudes

With the exponentials cancelled, the conditions Eq. 9.75 become Eq. 9.102:

$$\begin{gathered} \epsilon_1(\tilde{\mathbf{E}}_{0I} + \tilde{\mathbf{E}}_{0R})_z = \epsilon_2(\tilde{\mathbf{E}}_{0T})_z \\ (\tilde{\mathbf{B}}_{0I} + \tilde{\mathbf{B}}_{0R})_z = (\tilde{\mathbf{B}}_{0T})_z \\ (\tilde{\mathbf{E}}_{0I} + \tilde{\mathbf{E}}_{0R})_{x,y} = (\tilde{\mathbf{E}}_{0T})_{x,y} \\ \frac{1}{\mu_1}(\tilde{\mathbf{B}}_{0I} + \tilde{\mathbf{B}}_{0R})_{x,y} = \frac{1}{\mu_2}(\tilde{\mathbf{B}}_{0T})_{x,y} \end{gathered} \tag{9.102}$$

where $\tilde{\mathbf{B}}_0 = (1/v)\hat{k} \times \tilde{\mathbf{E}}_0$ for each wave.

5. Polarization parallel to the plane of incidence

Take the incident field in the plane of incidence (xz). By Prob. 9.15 the reflected and transmitted fields are in that plane too (Fig. 9.15). Condition (i) of Eq. 9.102 gives Eq. 9.103, condition (iii) gives Eq. 9.104, and condition (iv) gives Eq. 9.105:

$$\epsilon_1(-\tilde{E}_{0I}\sin\theta_I + \tilde{E}_{0R}\sin\theta_R) = \epsilon_2(-\tilde{E}_{0T}\sin\theta_T) \tag{9.103}$$

$$\tilde{E}_{0I}\cos\theta_I + \tilde{E}_{0R}\cos\theta_R = \tilde{E}_{0T}\cos\theta_T \tag{9.104}$$

$$\frac{1}{\mu_1 v_1}(\tilde{E}_{0I} - \tilde{E}_{0R}) = \frac{1}{\mu_2 v_2}\tilde{E}_{0T} \tag{9.105}$$

With the laws of reflection and refraction, Eq. 9.103 and Eq. 9.105 both reduce to

$$\tilde{E}_{0I} - \tilde{E}_{0R} = \beta \tilde{E}_{0T} \tag{9.106}$$

where

$$\beta \equiv \frac{\mu_1 v_1}{\mu_2 v_2} = \frac{\mu_1 n_2}{\mu_2 n_1} \tag{9.107}$$

and Eq. 9.104 becomes

$$\tilde{E}_{0I} + \tilde{E}_{0R} = \alpha \tilde{E}_{0T} \tag{9.108}$$

where

$$\alpha \equiv \frac{\cos\theta_T}{\cos\theta_I} \tag{9.109}$$

Solving for the amplitudes gives the Fresnel equations Eq. 9.110:

$$\begin{aligned} \tilde{E}_{0R} &= \left(\frac{\alpha - \beta}{\alpha + \beta}\right)\tilde{E}_{0I} \\ \tilde{E}_{0T} &= \left(\frac{2}{\alpha + \beta}\right)\tilde{E}_{0I} \end{aligned} \tag{9.110}$$

Here α depends on the angle of incidence through Snell's law:

$$\begin{aligned} \alpha &= \frac{\sqrt{1 - \sin^2\theta_T}}{\cos\theta_I} \\ &= \frac{\sqrt{1 - [(n_1/n_2)\sin\theta_I]^2}}{\cos\theta_I} \end{aligned} \tag{9.111}$$

At normal incidence $\alpha = 1$, which recovers Eq. 9.83. At grazing incidence α diverges, and the wave is totally reflected.

6. Brewster's angle

There is an angle θ_B at which the reflected wave vanishes. From Eq. 9.110, this is $\alpha = \beta$:

$$\sin^2\theta_B = \frac{1 - \beta^2}{(n_1/n_2)^2 - \beta^2} \tag{9.112}$$

For $\mu_1 \approx \mu_2$, $\beta \approx n_2/n_1$, and

$$\tan\theta_B \approx \frac{n_2}{n_1} \tag{9.113}$$

Footnote 19: at Brewster's angle the reflected beam is polarized perpendicular to the plane of incidence, so it is parallel to the surface, which is why polarizing sunglasses reduce glare from horizontal surfaces.

7. Intensities and coefficients

The power per unit area striking the interface is $\mathbf{S}\cdot\hat{z}$, so the cosine factors enter:

$$I_I = \frac{1}{2}\epsilon_1 v_1 E_{0I}^2 \cos\theta_I \tag{9.114}$$

and, for the reflected and transmitted waves,

$$\begin{aligned} I_R &= \frac{1}{2}\epsilon_1 v_1 E_{0R}^2 \cos\theta_R \\ I_T &= \frac{1}{2}\epsilon_2 v_2 E_{0T}^2 \cos\theta_T \end{aligned} \tag{9.115}$$

For polarization parallel to the plane of incidence:

$$R = \left(\frac{\alpha - \beta}{\alpha + \beta}\right)^2 \tag{9.116}$$

$$T = \alpha\beta\left(\frac{2}{\alpha + \beta}\right)^2 \tag{9.117}$$

Energy conservation gives $I_I = I_T + I_R$ Eq. 9.118, so $R + T = 1$. At Brewster's angle $R = 0$ and $T = 1$.

Problems 9.16–9.19 (a proof on exponentials, the polarization perpendicular to the plane, diamond, and a check on the Poynting vectors) are listed in the book but are not worked here.

Section 5, background beyond the syllabus list: absorption, dispersion and guided waves (Griffiths 9.4, 9.5)

9.4 Absorption and dispersion (background)

*Source: book-sources/m3-sec4-oblique.html (the first sheet, p. 418), m3-bg-absorption.html, Griffiths pp. 418–430. **Background**: beyond the syllabus list. Equations 9.119–9.176 are given in full. The derivations are in the book; these notes give the steps that lead to each result and the results themselves. Problems 9.20–9.27 are listed, not worked.*

Before you start

- Complex numbers, $\tilde{k} = k + i\kappa$, and the real part (Section 9.1.2).
- Ohm's law $\mathbf{J} = \sigma\mathbf{E}$ (Module 1 and Plus Two current electricity).
- Binomial expansion $\sqrt{1+\varepsilon} \approx 1 + \frac{1}{2}\varepsilon$ (pre-math M2).

1. Conductors (§9.4.1)

In a conductor the free current is not zero. With Ohm's law,

$$\mathbf{J}_f = \sigma\mathbf{E} \tag{9.119}$$

Maxwell's equations for a linear conductor take the form Eq. 9.120, after using ρ_f and Gauss's law:

$$\begin{aligned} &\text{(i)}\ \nabla\cdot\mathbf{E} = \frac{1}{\epsilon}\rho_f \\ &\text{(iii)}\ \nabla\times\mathbf{E} = -\frac{\partial\mathbf{B}}{\partial t} \\ &\text{(ii)}\ \nabla\cdot\mathbf{B} = 0 \\ &\text{(iv)}\ \nabla\times\mathbf{B} = \mu\sigma\mathbf{E} + \mu\epsilon\frac{\partial\mathbf{E}}{\partial t} \end{aligned} \tag{9.120}$$

The continuity equation for free charge is $\nabla\cdot\mathbf{J}_f = -\partial\rho_f/\partial t$ Eq. 9.121. With Ohm's law and Gauss's law it gives $\rho_f(t) = e^{-(\sigma/\epsilon)t}\rho_f(0)$ Eq. 9.122. So any free charge disappears in a time $\tau = \epsilon/\sigma$, and once it has gone, $\rho_f = 0$. Footnote 20 explains why τ for good conductors is not the real timescale.

With $\rho_f = 0$ the equations reduce to Eq. 9.123, and the extra term in (iv) is the only difference from the non-conducting case:

$$\begin{aligned} &\text{(i)}\ \nabla\cdot\mathbf{E} = 0 \\ &\text{(iii)}\ \nabla\times\mathbf{E} = -\frac{\partial\mathbf{B}}{\partial t} \\ &\text{(ii)}\ \nabla\cdot\mathbf{B} = 0 \\ &\text{(iv)}\ \nabla\times\mathbf{B} = \mu\epsilon\frac{\partial\mathbf{E}}{\partial t} + \mu\sigma\mathbf{E} \end{aligned} \tag{9.123}$$

Applying the curl to (iii) and (iv) gives modified wave equations Eq. 9.124:

$$\begin{aligned} \nabla^2\mathbf{E} &= \mu\epsilon\frac{\partial^2\mathbf{E}}{\partial t^2} + \mu\sigma\frac{\partial\mathbf{E}}{\partial t} \\ \nabla^2\mathbf{B} &= \mu\epsilon\frac{\partial^2\mathbf{B}}{\partial t^2} + \mu\sigma\frac{\partial\mathbf{B}}{\partial t} \end{aligned} \tag{9.124}$$

Plane waves $\tilde{\mathbf{E}} = \tilde{\mathbf{E}}_0 e^{i(\tilde{k}z-\omega t)}$ Eq. 9.125 solve these, provided

$$\tilde{k}^2 = \mu\epsilon\omega^2 + i\mu\sigma\omega \tag{9.126}$$

Writing $\tilde{k} = k + i\kappa$ Eq. 9.127, the real and imaginary parts are Eq. 9.128:

$$\begin{aligned} k &\equiv \omega\sqrt{\frac{\epsilon\mu}{2}}\left[\sqrt{1+\left(\frac{\sigma}{\epsilon\omega}\right)^2}+1\right]^{1/2} \\ \kappa &\equiv \omega\sqrt{\frac{\epsilon\mu}{2}}\left[\sqrt{1+\left(\frac{\sigma}{\epsilon\omega}\right)^2}-1\right]^{1/2} \end{aligned} \tag{9.128}$$

Check: for $\sigma = 0$, $\kappa = 0$ and $k = \omega\sqrt{\epsilon\mu}$, the vacuum-type result of §9.3.1.

The imaginary part gives attenuation Eq. 9.129:

$$\begin{aligned} \tilde{\mathbf{E}}(z,t) &= \tilde{\mathbf{E}}_0 e^{-\kappa z}e^{i(kz-\omega t)} \\ \tilde{\mathbf{B}}(z,t) &= \tilde{\mathbf{B}}_0 e^{-\kappa z}e^{i(kz-\omega t)} \end{aligned} \tag{9.129}$$

The **skin depth** is $d = 1/\kappa$ Eq. 9.130, the distance over which the amplitude falls by $1/e$. The wavelength, speed and index are Eq. 9.131:

$$\begin{aligned} \lambda &= \frac{2\pi}{k} \\ v &= \frac{\omega}{k} \\ n &= \frac{ck}{\omega} \end{aligned} \tag{9.131}$$

Maxwell's equations fix the polarization: $\mathbf{E}$ along x Eq. 9.132, and $\mathbf{B}$ along y with amplitude $\tilde{k}/\omega$ times

that of $\mathbf{E}$ Eq. 9.133. Writing $\tilde{k} = Ke^{i\phi}$ Eq. 9.134 with

$$\begin{aligned} K \equiv |\tilde{k}| &= \sqrt{k^2 + \kappa^2} \\ &= \omega\sqrt{\epsilon\mu\sqrt{1 + \left(\frac{\sigma}{\epsilon\omega}\right)^2}} \end{aligned} \tag{9.135}$$

$$\phi \equiv \tan^{-1}(\kappa/k) \tag{9.136}$$

the complex amplitudes are related by $B_0 e^{i\delta_B} = (Ke^{i\phi}/\omega)E_0 e^{i\delta_E}$ Eq. 9.137. Hence the magnetic field lags the electric field by ϕ, and

$$\delta_B - \delta_E = \phi \tag{9.138}$$

$$\begin{aligned} \frac{B_0}{E_0} &= \frac{K}{\omega} \\ &= \sqrt{\epsilon\mu\sqrt{1 + \left(\frac{\sigma}{\epsilon\omega}\right)^2}} \end{aligned} \tag{9.139}$$

$$\begin{aligned} \mathbf{E}(z,t) &= E_0 e^{-\kappa z}\cos(kz - \omega t + \delta_E)\hat{x} \\ \mathbf{B}(z,t) &= B_0 e^{-\kappa z}\cos(kz - \omega t + \delta_E + \phi)\hat{y} \end{aligned} \tag{9.140}$$

Figure 9.18 shows these fields. Problems 9.20–9.22 (skin depth in water and in metal, the phase lag, and the energy density) are listed, not worked.

2. Reflection at a conducting surface (§9.4.2)

At a surface with free charges and currents the boundary conditions are Eq. 9.141:

$$\begin{aligned} &\text{(i)}\ \epsilon_1 E_1^{\perp} - \epsilon_2 E_2^{\perp} = \sigma_f \\ &\text{(iii)}\ \mathbf{E}_1^{\parallel} - \mathbf{E}_2^{\parallel} = 0 \\ &\text{(ii)}\ B_1^{\perp} - B_2^{\perp} = 0 \\ &\text{(iv)}\ \frac{1}{\mu_1}\mathbf{B}_1^{\parallel} - \frac{1}{\mu_2}\mathbf{B}_2^{\parallel} = \mathbf{K}_f \times \hat{\mathbf{n}} \end{aligned} \tag{9.141}$$

For an ohmic conductor there is no free surface current (footnote 21). Take the incident, reflected and transmitted waves Eq. 9.142–Eq. 9.144, with the transmitted wave using $\tilde{k}_2$:

$$\begin{aligned} \tilde{\mathbf{E}}_I &= \tilde{E}_{0I} e^{i(k_1 z - \omega t)}\hat{x} \\ \tilde{\mathbf{B}}_I &= \frac{1}{v_1}\tilde{E}_{0I} e^{i(k_1 z - \omega t)}\hat{y} \end{aligned} \tag{9.142}$$

$$\begin{aligned} \tilde{\mathbf{E}}_R &= \tilde{E}_{0R} e^{i(-k_1 z - \omega t)}\hat{x} \\ \tilde{\mathbf{B}}_R &= -\frac{1}{v_1}\tilde{E}_{0R} e^{i(-k_1 z - \omega t)}\hat{y} \end{aligned} \tag{9.143}$$

$$\begin{aligned} \tilde{\mathbf{E}}_T &= \tilde{E}_{0T} e^{i(\tilde{k}_2 z - \omega t)}\hat{x} \\ \tilde{\mathbf{B}}_T &= \frac{\tilde{k}_2}{\omega}\tilde{E}_{0T} e^{i(\tilde{k}_2 z - \omega t)}\hat{y} \end{aligned} \tag{9.144}$$

Conditions (iii) and (iv) at $z = 0$ give Eq. 9.145 and Eq. 9.146, and then Eq. 9.147 with $\tilde{\beta}$ defined in Eq. 9.148:

$$\tilde{E}_{0I} + \tilde{E}_{0R} = \tilde{E}_{0T} \tag{9.145}$$

$$\frac{1}{\mu_1 v_1}(\tilde{E}_{0I} - \tilde{E}_{0R}) - \frac{\tilde{k}_2}{\mu_2\omega}\tilde{E}_{0T} = 0 \tag{9.146}$$

$$\begin{aligned} \tilde{E}_{0I} - \tilde{E}_{0R} &= \tilde{\beta}\tilde{E}_{0T} \\ \tilde{\beta} &\equiv \frac{\mu_1 v_1}{\mu_2\omega}\tilde{k}_2 \end{aligned} \tag{9.147}$$

Eq. 9.147 has the form of Eq. 9.81 with a complex $\tilde{\beta}$. Solving:

$$\begin{aligned} \tilde{E}_{0R} &= \left(\frac{1 - \tilde{\beta}}{1 + \tilde{\beta}}\right)\tilde{E}_{0I} \\ \tilde{E}_{0T} &= \left(\frac{2}{1 + \tilde{\beta}}\right)\tilde{E}_{0I} \end{aligned} \tag{9.149}$$

For a perfect conductor, $\tilde{k}_2 \to \infty$ and $\tilde{\beta} \to \infty$, so $\tilde{E}_{0R} = -\tilde{E}_{0I}$ and $\tilde{E}_{0T} = 0$ Eq. 9.150. The wave is totally reflected with a $180°$ phase shift; this is why good conductors make mirrors. Problem 9.23 (reflection from silver) is listed, not worked.

3. Frequency dependence of permittivity (§9.4.3)

Dispersion means the speed depends on frequency. The phase velocity is $v = \omega/k$ Eq. 9.151, and the group velocity of a packet is $v_g = d\omega/dk$ Eq. 9.152. The group velocity is the speed of the envelope and of the energy. Footnotes 23 and 24 give references, and the book notes that v_g can exceed c in special cases.

Model: each electron in a molecule is a damped oscillator with spring force $-m\omega_0^2 x$ Eq. 9.153 and damping force $-m\gamma\, dx/dt$ Eq. 9.154. Under the wave's field $E_0\cos\omega t$, the driving force is

$$F_{\text{driving}} = qE = qE_0\cos(\omega t) \tag{9.155}$$

and the electron obeys Eq. 9.156. In complex form Eq. 9.157, the steady state is $\tilde{x} = \tilde{x}_0 e^{-i\omega t}$ Eq. 9.158, so

$$\tilde{x}_0 = \frac{q/m}{\omega_0^2 - \omega^2 - i\gamma\omega}E_0 \tag{9.159}$$

The dipole moment is $\tilde{p} = q\tilde{x}$ Eq. 9.160. Summing over f_j electrons per molecule at frequency ω_j and damping γ_j, with N molecules per unit volume, gives

the polarization Eq. 9.161, the complex susceptibility Eq. 9.162, and

$$\tilde{\epsilon}_r = 1 + \frac{Nq^2}{m\epsilon_0}\sum_j \frac{f_j}{\omega_j^2 - \omega^2 - i\gamma_j\omega} \tag{9.163}$$

The complex wave equation Eq. 9.164 has the plane-wave solution Eq. 9.165 with $\tilde{k} = \sqrt{\tilde{\epsilon}\mu_0}\,\omega$ Eq. 9.166, and writing $\tilde{k} = k + i\kappa$ Eq. 9.167 gives the attenuated wave Eq. 9.168. The absorption coefficient is $\alpha = 2\kappa$ Eq. 9.169, and $n = ck/\omega$ Eq. 9.170.

For a dilute gas, using $\sqrt{1+\varepsilon} \approx 1 + \frac{1}{2}\varepsilon$, the complex wave number is Eq. 9.171, and the real and imaginary parts give Eq. 9.172 and Eq. 9.173:

$$n \approx 1 + \frac{Nq^2}{2m\epsilon_0}\sum_j \frac{f_j(\omega_j^2 - \omega^2)}{(\omega_j^2 - \omega^2)^2 + \gamma_j^2\omega^2} \tag{9.172}$$

$$\alpha \approx \frac{Nq^2\omega^2}{m\epsilon_0 c}\sum_j \frac{f_j\gamma_j}{(\omega_j^2 - \omega^2)^2 + \gamma_j^2\omega^2} \tag{9.173}$$

Away from resonance, n rises with frequency (normal dispersion). Near a resonance n falls sharply (anomalous dispersion), and the absorption is largest there (Fig. 9.22). Far from resonance, with damping ignored, $n = 1 + \frac{Nq^2}{2m\epsilon_0}\sum_j f_j/(\omega_j^2 - \omega^2)$ Eq. 9.174. For $\omega < \omega_j$ this becomes Eq. 9.175, and in terms of the vacuum wavelength, Cauchy's formula Eq. 9.176: $n = 1 + A(1 + B/\lambda^2)$.

Problems 9.24–9.27 (wave velocity and group velocity, hydrogen, the width of the anomalous region, a graph) are listed, not worked.

9.5 Guided waves (background)

*Source: book-sources/m3-bg-guided.html (and the start on p. 430 in m3-bg-absorption.html), Griffiths pp. 430–441. **Background**, beyond the syllabus list. Equations 9.177–9.206 in full, with the "More Problems on Chapter 9" equations (9.198–9.208) where they are used. Problems 9.28–9.44 are listed, not worked. Chapter 9 ends on p. 441; Chapter 10 starts on the next sheet and is not included.*

Before you start

- Maxwell's equations in vacuum (Eq. 9.40) and the vector identities in pre-math M13.
- Boundary conditions for a perfect conductor (Eq. 9.141 with no free surface current in the interior).
- Separation of variables for a partial differential equation (Plus Two has none; read the steps carefully).

1. Wave guides (§9.5.1)

A hollow pipe with a perfectly conducting wall is a wave guide (Fig. 9.23). Inside the metal $\mathbf{E} = \mathbf{0}$ and $\mathbf{B} = \mathbf{0}$. At the inner wall the boundary conditions are

$$\text{(i) } \mathbf{E}^{\parallel} = \mathbf{0}, \qquad \text{(ii) } B^{\perp} = 0 \tag{9.177}$$

Free charges and currents on the surface adjust to enforce these (footnote 27). For a monochromatic wave down the tube the fields have the form Eq. 9.178:

$$\begin{aligned}\text{(i)}\ \tilde{\mathbf{E}}(x,y,z,t) &= \tilde{\mathbf{E}}_0(x,y)e^{i(kz-\omega t)}\\ \text{(ii)}\ \tilde{\mathbf{B}}(x,y,z,t) &= \tilde{\mathbf{B}}_0(x,y)e^{i(kz-\omega t)}\end{aligned} \tag{9.178}$$

with k real. Maxwell's equations in the interior are Eq. 9.179:

$$\begin{aligned}&\text{(i)}\ \nabla\cdot\mathbf{E}=0\\ &\text{(iii)}\ \nabla\times\mathbf{E}=-\frac{\partial\mathbf{B}}{\partial t}\\ &\text{(ii)}\ \nabla\cdot\mathbf{B}=0\\ &\text{(iv)}\ \nabla\times\mathbf{B}=\frac{1}{c^2}\frac{\partial\mathbf{E}}{\partial t}\end{aligned} \tag{9.179}$$

Confined waves are in general *not* transverse. To satisfy Eq. 9.177 the field needs longitudinal components Eq. 9.180:

$$\begin{aligned}\tilde{\mathbf{E}}_0 &= E_x\hat{x}+E_y\hat{y}+E_z\hat{z}\\ \tilde{\mathbf{B}}_0 &= B_x\hat{x}+B_y\hat{y}+B_z\hat{z}\end{aligned} \tag{9.180}$$

Substituting into (iii) and (iv) gives six equations Eq. 9.181 for the transverse components in terms of the longitudinal ones. Solving four of them (Prob. 9.28a gives the details) yields Eq. 9.182:

$$\begin{aligned}\text{(i)}\ E_x &= \frac{i}{(\omega/c)^2-k^2}\left(k\frac{\partial E_z}{\partial x}+\omega\frac{\partial B_z}{\partial y}\right)\\ \text{(ii)}\ E_y &= \frac{i}{(\omega/c)^2-k^2}\left(k\frac{\partial E_z}{\partial y}-\omega\frac{\partial B_z}{\partial x}\right)\\ \text{(iii)}\ B_x &= \frac{i}{(\omega/c)^2-k^2}\left(k\frac{\partial B_z}{\partial x}-\frac{\omega}{c^2}\frac{\partial E_z}{\partial y}\right)\\ \text{(iv)}\ B_y &= \frac{i}{(\omega/c)^2-k^2}\left(k\frac{\partial B_z}{\partial y}+\frac{\omega}{c^2}\frac{\partial E_z}{\partial x}\right)\end{aligned} \tag{9.182}$$

So only E_z and B_z are needed; the rest follow by differentiation. Putting Eq. 9.182 back in the remaining Maxwell equations gives uncoupled equations Eq. 9.183:

$$\begin{aligned}\text{(i)}\ &\left[\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+(\omega/c)^2-k^2\right]E_z=0\\ \text{(ii)}\ &\left[\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+(\omega/c)^2-k^2\right]B_z=0\end{aligned} \tag{9.183}$$

If $E_z=0$ the wave is **TE** (transverse electric); if $B_z=0$ it is **TM** (transverse magnetic). If both are zero it is **TEM**. Hollow wave guides cannot carry TEM waves (the proof uses Gauss's and Faraday's laws for a field of zero divergence and zero curl, which must be the gradient of a potential that is constant on the wall; footnote 29).

2. TE modes in a rectangular guide (§9.5.2)

Separate variables in $B_z=X(x)Y(y)$. The equations are Eq. 9.184, with Eq. 9.185 linking the constants:

$$\begin{aligned}\text{(i)}\ \frac{1}{X}\frac{d^2X}{dx^2} &= -k_x^2\\ \text{(ii)}\ \frac{1}{Y}\frac{d^2Y}{dy^2} &= -k_y^2\end{aligned} \tag{9.184}$$

$$-k_x^2-k_y^2+(\omega/c)^2-k^2=0 \tag{9.185}$$

The boundary conditions (zero normal derivative at the walls) fix the constants:

$$k_x=\frac{m\pi}{a} \tag{9.186}$$

$$k_y=\frac{n\pi}{b} \tag{9.187}$$

so

$$B_z=B_0\cos(m\pi x/a)\cos(n\pi y/b) \tag{9.188}$$

This is the TE$_{mn}$ mode; by convention $a\geq b$. The wave number follows from Eq. 9.185:

$$k=\sqrt{(\omega/c)^2-\pi^2\left[(m/a)^2+(n/b)^2\right]} \tag{9.189}$$

If $\omega<\omega_{mn}$ the wave number is imaginary and the fields are attenuated (9.178 with imaginary k). The cutoff frequency is Eq. 9.190:

$$\omega_{mn}\equiv c\pi\sqrt{(m/a)^2+(n/b)^2} \tag{9.190}$$

The lowest cutoff is for TE$_{10}$: $\omega_{10}=c\pi/a$ Eq. 9.191. Writing k in terms of cutoff Eq. 9.192:

$$k=\frac{1}{c}\sqrt{\omega^2-\omega_{mn}^2} \tag{9.192}$$

The wave velocity $v=\omega/k$ is greater than c Eq. 9.193, but the energy travels at the group velocity Eq. 9.194, which is less than c:

$$v=\frac{\omega}{k}=\frac{c}{\sqrt{1-(\omega_{mn}/\omega)^2}} \tag{9.193}$$

$$v_g=\frac{1}{dk/d\omega}=c\sqrt{1-(\omega_{mn}/\omega)^2}<c \tag{9.194}$$

Figure 9.25 gives the picture: a plane wave bouncing between the walls at angle θ. Its wave vector is $\mathbf{k}'=(\pi m/a)\hat{x}+(\pi n/b)\hat{y}+k\hat{z}$, the frequency is $\omega=\sqrt{(ck)^2+\omega_{mn}^2}$ (the book's form of Eq. 9.190),

and $\cos\theta = k/|\mathbf{k}'| = \sqrt{1-(\omega_{mn}/\omega)^2}$. The plane wave travels at c but moves down the guide at $c\cos\theta = v_g$, while the wave fronts move at $c/\cos\theta = v$.

3. The coaxial line (§9.5.3)

A coaxial line has a central wire of radius a inside a conducting sheath of radius b. Unlike a hollow guide, it carries TEM waves: $E_z = 0$ and $B_z = 0$. Then Eq. 9.181 gives

$$k = \frac{\omega}{c} \tag{9.195}$$

(nondispersive), and

$$cB_y = E_x, \qquad cB_x = -E_y \tag{9.196}$$

with $\mathbf{E}$ and $\mathbf{B}$ perpendicular. The transverse fields obey the two-dimensional equations of electrostatics and magnetostatics Eq. 9.197:

$$\begin{aligned} \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} &= 0 \\ \frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} &= 0 \\ \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} &= 0 \\ \frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} &= 0 \end{aligned} \tag{9.197}$$

The cylindrically symmetric solution is borrowed from the line charge and the straight current Eq. 9.198:

$$\begin{aligned} \mathbf{E}_0(s,\phi) &= \frac{A}{s}\hat{s} \\ \mathbf{B}_0(s,\phi) &= \frac{A}{cs}\hat{\phi} \end{aligned} \tag{9.198}$$

Substituting and taking the real part gives the fields of the coaxial TEM wave Eq. 9.199:

$$\begin{aligned} \mathbf{E}(s,\phi,z,t) &= \frac{A\cos(kz-\omega t)}{s}\hat{s} \\ \mathbf{B}(s,\phi,z,t) &= \frac{A\cos(kz-\omega t)}{cs}\hat{\phi} \end{aligned} \tag{9.199}$$

Problem 9.33 asks for the charge and current on the inner wire.

4. Chapter-end problems (More Problems on Chapter 9)

The book ends the chapter with Problems 9.29–9.44. Their equations are copied in the book source. The statements used in the notes are: the inversion theorem for Fourier transforms (Prob. 9.34, Eq. 9.200); the transmission coefficient of a slab (Prob. 9.38, Eq. 9.201); the critical angle for total internal reflection (Prob. 9.41, Eq. 9.202), with the evanescent fields (Eqs. 9.203–9.205 in Prob. 9.41); the resonant frequencies of a closed box (Prob. 9.42, Eq. 9.206); and the optical-tweezers force (Prob. 9.44, Eqs. 9.207–9.208). None of these problems is worked here.

Problem 9.38: the transmission coefficient for normal incidence through a slab of medium 2 (Eq. 9.201) is given below.

$$T^{-1} = \frac{1}{4n_1n_3}\Big[(n_1+n_3)^2 + \frac{(n_1^2-n_2^2)(n_2^2-n_3^2)}{n_2^2}\sin^2\Big(\frac{n_2\omega d}{c}\Big)\Big] \tag{9.201}$$

Eq. 9.201 is used for its physical content: a slab can be made transparent at particular thicknesses. The critical angle Eq. 9.202 is

$$\theta_c \equiv \sin^{-1}(n_2/n_1) \tag{9.202}$$

and the evanescent fields of Problem 9.41 are $\tilde{\mathbf{E}}_T = \tilde{\mathbf{E}}_{0T}e^{-\kappa z}e^{i(kx-\omega t)}$ Eq. 9.203, with κ and k from Eq. 9.204.

Problem 9.42 (resonant cavity):

$$\omega_{lmn} = c\pi\sqrt{(l/d)^2+(m/a)^2+(n/b)^2} \tag{9.206}$$

Problem 9.44 (optical tweezers) gives the force Eq. 9.207, and with the averaged term dropped Eq. 9.208, $\mathbf{F} = \frac{\alpha}{2}\nabla E^2$.

The evanescent fields for polarization perpendicular to the plane of incidence are given by Eq. 9.205 in Problem 9.41(d); they are listed here as background.